EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Photo Electric Effect

1. Two monochromatic light sources, A and B, emit the same number of photons per second. The wavelength of A is $\lambda_A = 400\ nm$, and that of B is $\lambda_B = 600\ nm$. The power radiated by source B is

(A) equal to that of source A

(B) less than that of source A

(C) greater than that of source A

(D) cannot be compared to that from source A using the available data.

PMP089

2. When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is :

(A) 1 : 2 (B) 1 : 4 (C) 4 : 1 (D) 2 : 1

PMP103

3. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Plank's constant, c = Speed of light)

(A) $\frac{hc}{3\lambda}$ (B) $\frac{hc}{2\lambda}$ (C) $\frac{hc}{\lambda}$ (D) $\frac{2hc}{\lambda}$

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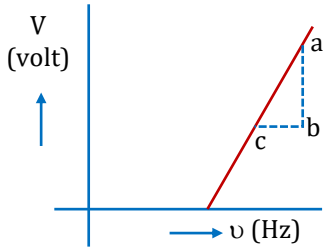
PMP105

4. Photons with energy 5 eV are incident on a cathode C in a photoelectric cell. The maximum energy of emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A, if the stopping potential of A relative to C is :

(A) – 1 V (B) – 3 V (C) + 3 V (D) + 4 V

PMP106

5. In a photoelectric experiment the graph of frequency u of incident light (in Hz) and stopping potential V (in volt) is shown below. From figure the value of the Planck's constant is (e is the elementary charge)

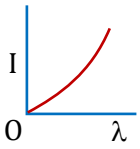


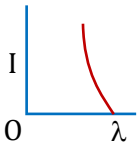
(A) $e\frac{ab}{cd}$ (B) $e\frac{cb}{ab}$ (C) $e\frac{ac}{bc}$ (D) $e\frac{ac}{ab}$

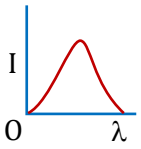
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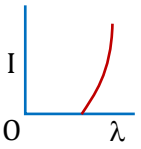
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6. The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current *I* of the photocell varies as follows :

(A) 

(B) 

(C) 

(D) 

PMP108

7. When a monochromatic point-source of light is at a distance of 0.2 *m* from a small photoelectric cell, the stopping potential and the saturation current are respectively 0.6 volt and 18.0 *mA*. If the same source is placed 0.6 *m* away from the photoelectric cell, then :-

(A) The stopping potential will be 0.2 volt (B) The stopping potential will be 1.8 volt

(C) The saturation current will be 6.0 *mA* (D) The saturation current will be 2.0 *mA*

PMP116

8. A silver sphere (work function 4.6 *eV*) is suspended in a vacuum chamber by an insulating thread. Ultraviolet light of wavelength 0.2 μ *m* strikes on the sphere. The maximum electric potential of the sphere will be (hc = 12400 *eVÅ*):

(A) 4.6 *V* (B) 6.2 *V* (C) 1.6 *V* (D) 3.2 *V*

PMP119

9. Light coming from a discharge tube filled with hydrogen falls on the cathode of the photoelectric cell. The work function of the surface of cathode is 4*eV*. Which one of the following values of the anode voltage (in Volts) with respect to the cathode will likely to make the photo current zero.

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(A) – 4 (B) – 6 (C) – 8 (D) – 10

PMP150

10. If light of wavelength of maximum intensity emitted from surface at temperature T_1 is used to cause photoelectric emission from a metallic surface, the maximum kinetic energy of the emitted electron is 6 *eV*, which is 3 times the work function of the metallic surface. If light of wavelength of maximum intensity emitted from a surface at temperature T_2 ($T_2 = 2T_1$) is used, the maximum kinetic energy of the photoelectrons emitted is :-

(A) 2 *eV* (B) 4 *eV* (C) 14 *eV* (D) 18 *eV*

PMP159

Atomic Structure

11. A hydrogen like gas atoms absorb radiations of wavelength λ_0 and consequently emit radiations of 6 difference wavelengths of which, three wavelengths are shorter than λ_0. Choose the correct alternative(*s*):

(A) The final excited state of the atoms is $n = 3$.

(B) The final excited state of the atoms is $n = 4$.

(C) The initial state of the atoms is $n = 1$.

(D) The initial state of the atoms is $n = 3$.

PMP120

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12. The de-Broglie wavelength of an electron in the first Bohr orbit is

(A) equal to the circumference of first orbit (B) equal to $\frac{1}{2}$ × (circumference of first orbit)

(C) equal to $\frac{1}{4}$ × (circumference of first orbit) (D) equal to $\frac{3}{4}$ × (circumference of first orbit)

PMP121

13. According to Bohr model, magnetic field at the centre (at the nucleus) of a hydrogen atom due to the motion of the electron in nth orbit is proportional to

(A) $1/n^3$ (B) $1/n^5$ (C) n^5 (D) n^3

PMP122

14. Consider 3rd orbit of He^+ (Helium), using non-relativistic approach, the speed of electron in this orbit will be [given $K = 9 \times 10^9$ constant, $Z = 2$ and h (Planck's Constant) = 6.6×10^{-34} Js]

(A) 1.46×10^6 m/s (B) 0.73×10^6 m/s (C) 3.0×10^8 m/s (D) 2.92×10^6 m/s

PMP123

15. In a hydrogen atom, the electron is in nth excited state. It may come down to second excited state by emitting ten different wavelengths. What is the value of n:

(A) 6 (B) 7 (C) 8 (D) 5

PMP125

16. The radius of the first permitted Bohr orbit for the electron, in a hydrogen atom equals 0.51 Å and its ground state energy equals –13.6 eV. If the electron in the hydrogen atom is replaced by muon [charge same as electron and mass 207 m_e], the first Bohr radius and ground state energy will be :

(A) 0.53×10^{-13} m, –3.6 eV (B) 25.6×10^{-13} m, –2.8 eV

(C) 2.56×10^{-13} m, –2.8 keV (D) 2.56×10^{-13} m, –13.6 eV

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PMP126

17. In the spectrum of single ionised helium, the wavelength of a line observed is almost the same as the first line of Balmer series of hydrogen. It is due to transition of electron from

(A) $n_1 = 6$ to $n_2 = 4$ (B) $n_1 = 5$ to $n_2 = 3$ (C) $n_1 = 4$ to $n_2 = 2$ (D) $n_1 = 3$ to $n_2 = 2$

PMP127

18. If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where a = _____.

(A) 144 (B) 145 (C) 245 (D) 241

PMP305

19. A photon of 10.2 eV energy collides with a hydrogen atom in ground state inelastically. After few microseconds one more photon of energy 15 eV collides with the same hydrogen atom. Then what can be detected by a suitable detector.

(A) one photon of 10.2 eV and an electron of energy 1.4 eV

(B) 2 photons of energy 10.2 eV

(C) 2 photons of energy 3.4 eV

(D) 1 photon of 3.4 eV and one electron of 1.4 eV

PMP131

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20. Energy levels A, B and C of a certain atom correspond to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are wave lengths of radiations corresponding to transitions C to B, B to A and C to A respectively, which of the following relations is correct :

(A) $\lambda_3 = \lambda_1 + \lambda_2$ (B) $\lambda_3 = \dfrac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$ (C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (D) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

PMP132

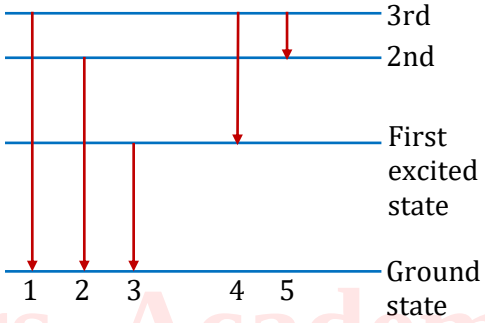
21. The ionization energy of the electron in the hydrogen atom in its ground state is 13.6 eV. The atoms are excited to higher energy levels to emit radiations of 6 wavelengths. Maximum wavelength of emitted radiation corresponds to the transition between :

(A) $n = 4$ to $n = 3$ states (B) $n = 3$ to $n = 2$ states

(C) $n = 3$ to $n = 1$ states (D) $n = 2$ to $n = 1$ states

PMP133

22. The figure indicates the energy level diagram of an atom and the origin of five spectral lines in emission spectra. Which of the spectral lines will also occur in the absorption spectra?



(A) 1, 2, 4 (B) 2, 3, 4, 5 (C) 1, 2, 3 (D) 1, 2, 3, 4, 5

PMP137

23. A stationary hydrogen atom of mass M emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is

(R = Rydberg constant and h = Plank's constant)

(A) $\dfrac{Rh}{M}$ (B) $\dfrac{Rh}{4M}$ (C) $\dfrac{3}{4}\dfrac{Rh}{M}$ (D) $\dfrac{5}{36}\dfrac{Rh}{M}$

PMP146

24. A force of attraction between the positively charged nucleus and the negatively charged electron in the hydrogen atom is given by $F = \dfrac{ke^2}{r^2}$ where k is the constant. The electron, initially moving in a circle of radius R_1 about the nucleus, jumps suddenly into a circular orbit of radius R_2. The total energy of the atom decreased in this process is :-

(A) $ke^2\left[\dfrac{1}{R_1}+\dfrac{1}{R_2}\right]$ (B) $\dfrac{ke^2}{2}\left[\dfrac{1}{R_2}-\dfrac{1}{R_1}\right]$ (C) $ke^2\left[\dfrac{R_1R_2}{R_2-R_1}\right]$ (D) $ke^2\left[\dfrac{R_1R_2}{R_1+R_2}\right]$

PMP148

25. The attractive potential for an atom is given by $v = v_0 \ln(r/r_0)$, v_0 and r_0 are constant and r is the radius of the orbit. The radius r of the nth Bohr's orbit depends upon principal quantum number n as:

(A) $r \propto n$ (B) $r \propto 1/n^2$ (C) $r \propto n^2$ (D) $r \propto 1/n$

PMP149

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X-rays

26. The *X*-ray beam coming from an *X*-ray tube will be

(A) monochromatic

(B) having all wavelengths smaller than a certain maximum wavelength

(C) having all wavelengths larger than a certain minimum wavelength

(D) having all wavelengths lying between a minimum and a maximum wavelength

PMP161

27. In an *X*-ray experiment target is made up of copper ($Z = 29$) having some impurity. The K_α line of copper have wavelength λ_0. It was observed that another K_α line due to impurity have wavelength $\frac{784}{625}\lambda_0$. The atomic number of the impurity element is

(A) 22 (B) 23 (C) 24 (D) 26

PMP163

28. Following statements are given for atom in ground state. Choose CORRECT statement :-

(A) Helium (*He*) can not emit K_β characteristic *X*-rays but can emit K_α characteristic *X*-rays

(B) Helium (*He*) can emit K_α as well as K_β characteristic *X*-rays

(C) Lithium (*Li*) can not emit K_β characteristic *X*-rays but can emit K_α characteristic *X*-rays

(D) Lithium (*Li*) can emit K_α as well as K_β characteristic *X*-rays

PMP164

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29. The energy of a *K*-electron in tungsten is –20 *keV* and of an *L*-electrons is –2 *keV*. The wave length of *X*-rays emitted when there is electron jump from *L* to *K* shell :

(A) 0.3443 Å (B) 0.6887 Å (C) 1.3982 Å (D) 2.78 Å

PMP165

30. The intensity of gamma radiation from a given source is *I*. On passing through 36 *mm* of lead, it is reduced to $I/8$. The thickness of lead, which will reduce the intensity to $I/2$ will be

(A) 6 *mm* (B) 9 *mm* (C) 18 *mm* (D) 12 *mm*

PMP166

31. Choose the INCORRECT statement.

(A) Cut-off wavelength of *X*-rays is independent of filament voltage.

(B) Presence of K_α-line in *X*-ray spectrum means that *L*-series will also be present.

(C) Increase in filament current increases intensity of *X*-ray.

(D) Presence of *L*-series in *X*-ray spectrum means that *K*-series will also be present.

PMP169

32. The *K*, *L* and *M* energy levels of platinum lie roughly at 78, 12 and 3 *keV* respectively. The ratio of wavelength of K_α line to that of K_β line in *X*-ray spectrum is-

(A) $\frac{22}{3}$ (B) $\frac{3}{22}$ (C) $\frac{22}{25}$ (D) $\frac{25}{22}$

PMP170

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33. Electrons with de-Broglie wavelength λ fall on the target in an *X*-ray tube. The cut-off wavelength of the emitted *X*-rays is

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$ (C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

PMP173

De Broglie

34. A free particle with initial kinetic energy *E* and de–broglie wavelength λ enters a region in which it has potential energy *U*. What is the particle's new de–Broglie wavelength?

(A) $\lambda(1 - U/E)^{-1/2}$ (B) $\lambda(1 - U/E)$ (C) $\lambda(1 - E/U)^{-1}$ (D) $\lambda(1 + U/E)^{1/2}$

PMP096

35. After absorbing a slowly moving neutron of mass m_N (momentum ~ 0) a nucleus of mass *M* breaks into two nuclei of masses m_1 and $3m_1(4m_1 = M + m_N)$, respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ, then de-Broglie wavelength of the other nucleus will be:-

(A) 9 λ (B) 3 λ (C) $\frac{\lambda}{3}$ (D) λ

PMP098

36. Two hydrogen atoms are in excited state with electrons residing in $n = 2$. First one is moving towards left and emits a photon of energy E_1 towards right. Second one is moving towards left with same speed and emits a photon of energy E_2 towards left. Taking recoil of nucleus into account during emission process:

(A) $E_1 > E_2$ (B) $E_1 < E_2$ (C) $E_1 = E_2$ (D) Information insufficient

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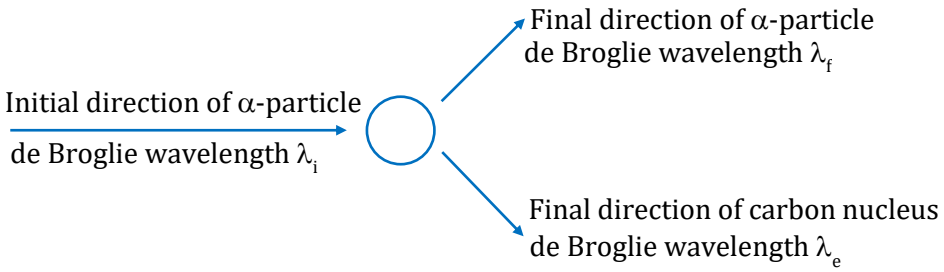
PMP141

37. A neutron having kinetic energy 13.6 *eV* collides with a hydrogen atom in ground state at rest. Assume that the mass of neutron and hydrogen atoms are same and the neutron does not leave its line of motion. Then which of the following is the possible *KE* of the neutron after the collision?

(A) Zero (B) 3.4 *eV* (C) 1.5 *eV* (D) 6.8 *eV*

PMP142

38. An α-particle having a de Broglie wavelength λ_i collides with a stationary carbon nucleus. The α-particle moves off in a different direction as shown below.



After the collision, the de Broglie wavelength of the α-particle and the carbon nucleus are λ_f and λ_e respectively. Which of the following relations about de Broglie wavelength is correct?

(A) $\lambda_i < \lambda_f$ (B) $\lambda_i > \lambda_f$ (C) $\lambda_f = \lambda_e$ (D) $\lambda_i = \lambda_e$

PMP145

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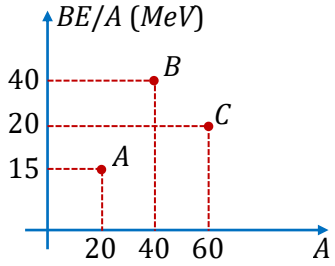
Nuclear Physics

39. Binding energy per nucleon of a fixed nucleus X^A is 8 *MeV*. It absorbs a neutron moving with kinetic energy 2 *MeV* and converts into *Y*, emitting two photon of energy 1 *MeV* and 4 *MeV* respectively one after the other. The binding energy of per nucleon of *Y* (in *MeV*) is

(A) $\frac{(8A-7)}{(A+1)}$ (B) $\frac{(8A+3)}{(A+1)}$ (C) $\frac{(8A+7)}{(A+1)}$ (D) $\frac{(8A-7)}{(A+1)}$

PMP178

40. The graph shows *BE/A* values for three hypothetical elements *A*, *B*, *C*. The energy released when *C* disintegrates into *A* and *B* is :

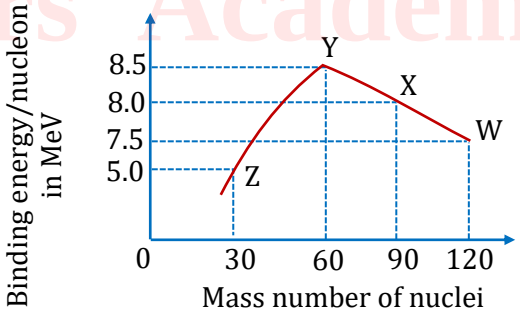


(A) 1000 *MeV* (B) 1200 *MeV* (C) 700 *MeV* (D) 500 *MeV*

PMP180

41. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. *W*, *X*, *Y* and *Z* are four nuclei indicated on the curve. The process that would release energy is :

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(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$ (C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

PMP181

42. The Binding energy per nucleon of ^7_3Li and ^4_2He nuclei are 5.60 *MeV* and 7.06 *MeV*, respectively. In the nuclear reaction $^7_3Li + {}^1_1H \rightarrow 2^4_2He + Q$, the value of energy *Q* released is :

(A) 19.6 *MeV* (B) –2.4 *MeV* (C) 8.4 *MeV* (D) 17.3 *MeV*

PMP184

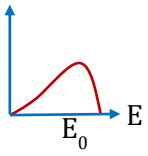
43. In the uranium radioactive series the initial nucleus is $_{92}U^{238}$, and the final nucleus is $_{82}Pb^{206}$. When the uranium nucleus decays to lead, the number of α – particles emitted is..... and the number of β-particles emitted is .....

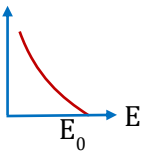
(A) 6, 8 (B) 8, 6 (C) 16, 6 (D) 32, 12

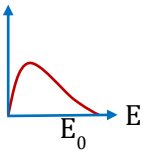
PMP185

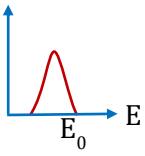
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44. The energy spectrum of β-particles, number $N(E)$ as a function of β-energy E emitted from a radioactive source is-

(A) 

(B) 

(C) 

(D) 

PMP187

45. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is :

(A) $E\left(^{236}_{92}U\right) > E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$ (B) $E\left(^{236}_{92}U\right) < E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$

(C) $E\left(^{236}_{92}U\right) < E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$ (D) $E\left(^{236}_{92}U\right) = E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$

PMP203

46. If each fission in a U^{235} nucleus releases 200 *MeV*, how many fissions must occurs per second to produce a power of 1 *KW*

(A) 1.325×10^{13} (B) 3.125×10^{13} (C) 1.235×10^{13} (D) 2.135×10^{13}

PMP208

Radioactivity

47. The decay constant of the end product of a radioactive series is

(A) zero (B) infinite

(C) finite (non zero) (D) depends on the end product.

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PMP189

48. Starting with a sample of pure 66 *Cu*, 7/8 of it decays into Z*n* in 15 *min*. The corresponding half-life is-

(A) 10 *min* (B) 15 *min* (C) 5 *min* (D) $7\frac{1}{2}$ *min*

PMP190

49. Two radioactive nuclei P and Q in a given sample decay into a stable nucleus R. At time $t = 0$, number of P species are 4 N_0 and that of Q are N_0. Half-life of P (for conversion to R) is 1 minute where as that of Q is 2 minutes. Initially there are no nuclei of R present in the sample. When number of nuclei of P and Q are equal, the number of nuclei of R present in the sample would be :

(A) $2N_0$ (B) $3N_0$ (C) $\frac{9N_0}{2}$ (D) $\frac{5N_0}{2}$

PMP191

50. A mixture consists of two radioactive materials A_1 and A_2 with half lives of 20s and 10*s* respectively. Initially the mixture has 40*g* of A_1 and 160g of A_2. The active amount of the two in the mixture will becomes equal after :

(A) 20*s* (B) 40*s* (C) 60*s* (D) 80*s*

PMP192

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51. A radioactive source in the form of a metal sphere of diameter 3.2×10^{-3} *m* emits *b*-particle at a constant rate of 6.25×10^{10} particle/sec. The source is electrically insulated and all the β-particle are emitted from the surface. The potential of the sphere will rise to 1 *V* in time

(A) 180 μ sec (B) 90 μ sec (C) 18 μ sec (D) 9 μ sec

PMP195

52. The half life of a radioactive isotope '*X*' is 20 years. It decays to another element '*Y*' which is stable. The two elements '*X*' and '*Y*' were found to be in the ratio 1 : 7 in a sample of a given rock. The age of the rock is estimated to be:

(A) 100 years (B) 40 years (C) 60 years (D) 80 years

PMP196

53. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it has decayed and time t_1 when $\frac{1}{3}$ of it had decayed is :

(A) 20 *min* (B) 28 *min* (C) 7 *min* (D) 14 *min*

PMP198

54. The decay constant of a radio isotope is λ. If A_1 and A_2 are its activities at times t_1 and t_2 respectively, the number of nuclei which have decayed during the time $(t_1 - t_2)$:

(A) $A_1 - A_2$ (B) $(A_1 - A_2)/\lambda$ (C) $\lambda(A_1 - A_2)$ (D) $A_1t_1 - A_2t_2$

PMP199

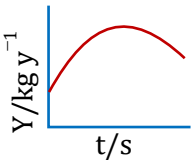
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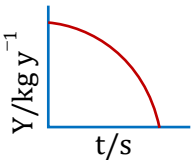
55. Half life of radium is 1620 years. How many radium nuclei decay in 5 hours in 5 *gm* radium? (Atomic weight of radium = 223)

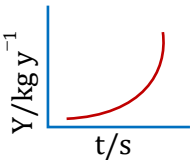
(A) 9.1×10^{12} (B) 3.23×10^{15} (C) 1.72×10^{20} (D) 3.3×10^{17}

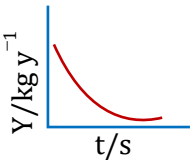
PMP200

56. The radioactive nucleus of an element *X* decays to a stable nucleus of element *Y*. A graph of the rate of formation of *Y* against time would look like

(A) 

(B) 

(C) 

(D) 

PMP201

57. A radioactive substance is dissolved in a liquid and the solution is heated. The activity of the solution

(A) is smaller than that of element

(B) is greater than that of element

(C) is equal to that of element

(D) will be smaller or greater depending upon whether the solution is weak or concentrated.

PMP204

For More Join: @IITJEE_Advanced

58. A radioactive sample S_1 having an activity of 5μ*Ci* has twice the number of nuclei as another sample S_2 which has an activity of 10μ*C*i. The half lives of S_1 and S_2 can be :

(A) 20 years and 5 years, respectively (B) 20 years and 10 years, respectively

(C) 10 years each (D) 5 years each

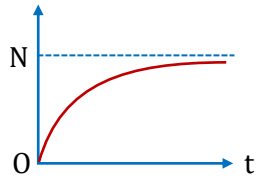
PMP210

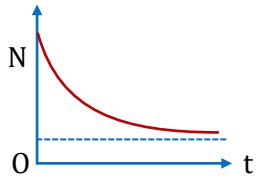
59. In a certain nuclear reactor, a radioactive nucleus is being produced at a constant rate = 1000 /*s*. The mean life of the radionuclide is 40 minutes. At steady state, the number of radionuclide will be

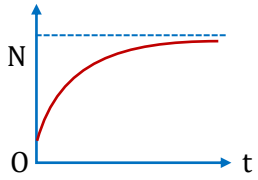
(A) 4×10^4 (B) 24×10^4 (C) 24×10^5 (D) 24×10^6

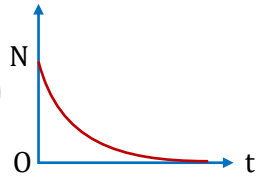
PMP211

60. In the above question, if there were 20×10^5 radionuclide at $t = 0$, then the graph of N *v/s* t is

(A) 

(B) 

(C) 

(D) 

PMP212

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61. The half-life of substance *X* is 45 years, and it decomposes to substance *Y*. A sample from a meteorite was taken which contained 2% of *X* and 14% of *Y* by quantity of substance. If substance *Y* is not normally found on a meteorite, what is the approximate age of the meteorite?

(A) 270 years (B) 135 years (C) 90 years (D) 45 years

PMP213

62. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals:

(A) $\frac{T}{\ln 2}\left|\ln\frac{A_1}{A_2}\right|$ (B) $T\left|\ln\frac{A_1}{A_2}\right|$ (C) $\frac{T}{\ln 2}\left|\ln\frac{A_2}{2A_1}\right|$ (D) $T\left|\ln\frac{A_2}{2A_1}\right|$

PMP216

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. A small plate of area 1 cm^2 is placed at a distance of $\frac{1}{\sqrt{\pi}}$ m from an isotropic point source emitting light of frequency $\frac{1}{6.63}\times10^{14}$ Hz, at a power of 2.00 mW. Assume the plate to be normal to the incident photons. [where $h = 6.63 \times 10^{-34}$ J-s]. Select CORRECT alternative(s):

(A) Energy possessed by each photon is 10^{-20} J

(B) Photon emission rate is 2×10^{17} s^{-1}

(C) The fraction of area of beam intercepted by the plate is $\frac{1}{4}\times10^{-4}$

(D) The rate of photons striking the plate is 5×10^{12} per second.

PMP091

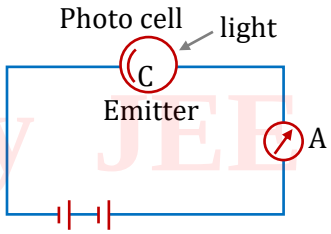
2. The figure shows a photo cell circuit. The cathode of the photo cell is illuminated by a monochromatic light. If the intensity is kept constant and the frequency of the incident light is increased, then the

(A) photo electric current in the circuit increases

(B) photo electric current in the circuit decreases

(C) maximum kinetic energy of the photo electrons increases

(D) photo electric current in the circuit can be reduced to zero, when the polarity of the terminals is reversed



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PMP109

3. Photoelectric effect supports quantum nature of light because

(A) there is minimum frequency of light below which no photoelectrons are emitted

(B) the maximum kinetic energy of photo-electrons depends only on the frequency of light and not on its intensity

(C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately

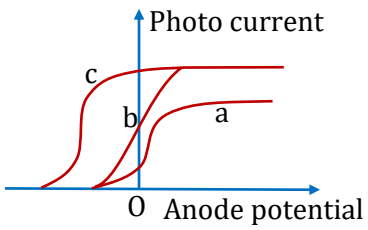
(D) electric charge of photo-electrons is quantized

PMP110

4. The figure shows the variation of photo current with anode potential for a photosensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves a, b and c respectively. Choose correct options:

(A) $f_a = f_b$ (B) $I_a < I_b$

(C) $f_c < f_b$ (D) $I_c > I_b$



PMP112

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5. Light of wavelength λ_1 and λ_2 are falling on two metal surface *A* and *B*. For wavelength λ_1 electron ejected from both the surfaces and for wavelength λ_2 electron ejected from only surface *B*. On the basis of these facts which one of the following is a false statement

(A) more energy is required for ejection of electron from metal '*A*'

(B) $\lambda_1 > \lambda_2$

(C) threshold wavelength for *A* is greater than *B*

(D) energy of electron ejected from metal *A* will be greater than electron ejected from metal *B* for wavelength λ

PMP113

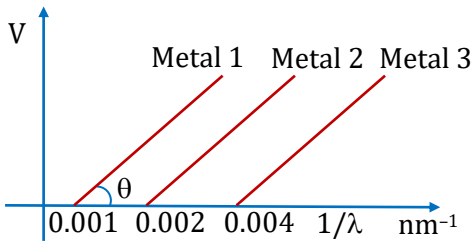
6. The graph between $1/\lambda$ and stopping potential (V) of three metals having work functions ϕ_1, ϕ_2 and ϕ_3 in an experiment of photo-electric effect is plotted as shown in the figure. Which of the following statement(*s*) is/are correct? [Here λ is the wavelength of the incident ray].

(A) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$

(B) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$

(C) $\tan\theta$ is directly proportional to hc/e, where h is Planck's constant and c is the speed of light

(D) The violet colour light can eject photoelectrons from metals 2 and 3.



PMP114

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7. The collector of the photocell (in photoelectric experiment) is made of tungsten while the emitter is of Platinum having work function of 10 e*V*. Monochromatic radiation of wavelength 124 *Å* and power 100 watt is incident on emitter which emits photo electrons with a quantum efficiency of 1%. The accelerating voltage across the photocell is of 10,000 volts. (Use : $hc = 12400 eV Å$)

(A) The power supplied by the accelerating voltage source is 100 watt

(B) The minimum wavelength of radiation coming from the tungsten target (collector) is 1.23 Å

(C) The power supplied by the accelerating voltage source is 10 watt

(D) The minimum wavelength of radiation coming from the tungsten target (collector) is 2.23 Å

PMP115

8. A metallic sphere of radius *r* remote from all other bodies is irradiated with a radiation of wavelength λ which is capable of causing photoelectric effect.

(A) the maximum potential gained by the sphere will be independent of its radius

(B) the net positive charge appearing on the sphere after a long time will depend on the radius of the sphere

(C) the maximum kinetic energy of the electrons emanating from the sphere will keep on declining with time

(D) the kinetic energy of the most energetic electrons emanating from the sphere initially will be independent of the radius of the sphere.

PMP118

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9. The accelerating potential (V) applied between a photocathode and the respective anode is such that the fastest photoelectron can fly only one fourth of the distance between the cathode and the anode. If the distance between photocathode and anode is reduced to $(1/4)^{th}$ of the original value while maintaining the accelerating potential constant, then

(A) The fastest electron will reach the anode

(B) The fastest electron will reach upto one fourth of the new distance between the cathode and the anode

(C) Kinetic energy of the fastest photoelectron emitted from photocathode will not change due to change in plate separation (keeping V constant)

(D) Kinetic energy of the fastest photoelectron will increase due to decrease in plate separation (keeping V constant)

PMP158

De Broglie

10. Two electrons are moving with the same speed v. One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field, then after sometime if the de–Broglie wavelengths of the two are λ_1 and λ_2, then select the possible option(s) :-

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 > \lambda_2$ (C) $\lambda_1 < \lambda_2$ (D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$

PMP099

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11. A particle moves in a closed orbit around the origin, due to a central force which is directed towards the origin. The de Broglie wavelength of the particle varies cyclically between two values λ_1, λ_2 with $\lambda_1 > \lambda_2$. Which of the following statements is/are true ?

(A) The particle could be moving in a circular orbit with origin as centre.

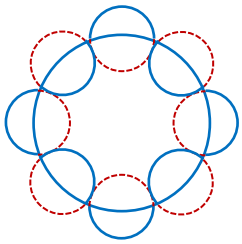
(B) The particle could be moving in an elliptic orbit with origin as its focus.

(C) When the de Broglie wave length is λ_1, the particle is nearer the origin than when its value is λ_2.

(D) When the de Broglie wavelength is λ_2, the particle is nearer the origin than when its value is λ_1.

PMP100

12. Figure shows stationary orbit of an hydrogen atom upon the transition of electron from given excited state to ground state.



(A) Average change in angular momentum is $\frac{3h}{2\pi}$.

(B) The ratio of de-Broglie wavelengths in final state to initial state is 4.

(C) Energy of emitted photon is nearly 12.75 eV.

(D) Ratio of orbital time period in final state to initial state is 4^3

PMP130

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13. A beam of ultraviolet light of all wavelengths passes through hydrogen gas at room temperature, in the *x*-direction. Assume that all photons emitted due to electron transition inside the gas emerge in the *y*-direction. Let *A* and *B* denote the lights emerging from the gas in the *x* and *y* directions respectively.

(A) Some of the incident wavelengths will be absent in *A*.

(B) Only those wavelengths will be present in *B* which are absent in *A*.

(C) *B* will contain some visible light.

(D) *B* will contain some infrared light.

PMP134

14. Energy liberated in the de-excitation of hydrogen atom from 3rd level to 1st level falls on a photo- cathode. Later when the same photocathode is exposed to a spectrum of some unknown hydrogen like gas, excited to 2nd energy level, it is found that the de-Broglie wavelength of the fastest photoelectrons, now ejected has decreased by a factor of 3. For this new gas, difference of energies of 2nd Lyman line and 1st Balmer line is found to be 3 times the ionization potential of the hydrogen atom. Select the correct statement(s):

(A) The gas is lithium

(B) The gas is helium

(C) The work function of photocathode is 8.5 *eV*

(D) The work function of photocathode is 5.5 *eV*

PMP136

15. A particular hydrogen like atom has its ground state binding "energy 122.4*eV* ". Its is in ground state. Then:

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(A) Its atomic number is 3.

(B) An electron of 90 *eV* can excite it.

(C) An electron of kinetic energy nearly 91.8 *eV* can be brought to almost rest by this atom.

(D) An electron of kinetic energy 2.6 *eV* may emerge from the atom when electron of kinetic energy 125 *eV* collides with this atom.

PMP140

16. A neutron collides head-on with a stationary hydrogen atom in ground state. Which of the following statements are correct. (Assume that the hydrogen atom and neutron has same mass) :

(A) If kinetic energy of the neutron is less than 20.4 *eV* collision must be elastic.

(B) If kinetic energy of the neutron is less than 20.4 *eV* collision may be inelastic.

(C) Inelastic collision may be take place only when initial kinetic energy of neutron is greater than 20.4 *eV*.

(D) Perfectly inelastic collision cannot take place.

PMP143

17. A free hydrogen atom in ground state is at rest. A neutron of kinetic energy '*K*' collides with the hydrogen atom. After collision hydrogen atom emits two photons in succession one of which has energy 2.55 *eV*. (Assume that the hydrogen atom and neutron has same mass)

(A) minimum value of '*K*' is 25.5 *eV*.

(B) minimum value of '*K*' is 12.75 *eV.*

(C) the other photon has energy 10.2 *eV* if *K* is minimum.

(D) the upper energy level is of excitation energy 12.75 *eV*.

PMP144

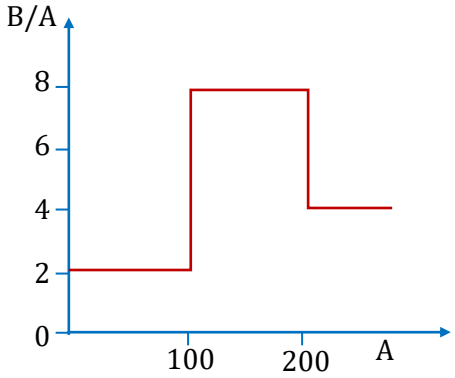
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18. Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mass of a ${}^{20}_{10}Ne$ nucleus and M_2 the mass of a ${}^{40}_{20}Ca$ nucleus. Then

(A) $M_2 = 2M_1$ (B) $M_2 > 2M_1$ (C) $M_2 < 2M_1$ (D) $M_1 < 10(m_n + m_p)$

PMP175

19. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below:



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy

(B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments

(D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

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PMP182

20. In β-decay, the Q-value of the process is E. Then

(A) $K.E.$ of a β-particle cannot exceed E.

(B) $K.E.$ of anti neutrino emitted lies between Zero and E.

(C) N/Z ratio of the nucleus is altered.

(D) Mass number (A) of the nucleus is altered.

PMP188

21. Consider the following nuclear reactions and select the correct statements from the options that follow:

Reaction I : $n \rightarrow p + e^- + \bar{\nu}$

Reaction II : $p \rightarrow n + e^+ + \nu$

(a) Free neutron has higher mass than proton, therefore reaction I is possible.

(b) Free proton has less mass than neutron, therefore reaction II is not possible for the free proton.

(c) Inside a nucleus, both decay (reaction I and II) are possible.

(d) Free proton is not stable outside nucleus.

(A) a, b, c are correct (B) a, c, d are correct

(C) a, b, d are correct (D) b, c, d are correct

PMP306

For More Join: @IITJEE_Advanced

EXERCISE (S)

Photo Electric Effect

1. A 5 watt source emits monochromatic light of wavelength 5000 Å. When placed 0.5 m away, it liberates photoelectrons from a photosensitive metallic surface. When the source is moved to a distance of 1.0 m, the number of photo electrons liberated will be reduced by a factor of :

PMP090

2. A cooling object was emitting radiations of time varying wavelength $\lambda = 3000 + 40t$, where λ is in Å and t is in second is incident on a metal sheet (of work function $2eV$) such that the power incident on sheet is constant at 100 watt. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, λ again starts from fresh value of 3000 Å. If the metal plate is grounded so that it always remains neutral and electron clouding is negligible then find the maximum photocurrent(mA). The photoemission efficiency is 0.01% and remains constant. (Take $hc = 12400\ eV$-Å)

PMP219

3. A light of wavelength 3540 Å falls on a metal having work function of 2.5 eV. If ejected electron collides with another target metal inelastically and its total kinetic energy is utilized to raise the temperature of target metal. The mass of target metal is $10^{-3}\ kg$ and its specific heat is $160\ J/kg/°C$. If 10^{18} electrons are ejected per second, then find the rate of raise of temperature (in $°C/s$) of the metal.

[Assume there is no loss of energy of ejected electron by any other process, all the electrons are reaching the target metal with max kinetic energy and take $hc = 12400\ ev$-Å]

PMP220

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Atomic Structure

4. A neutron beam, in which each neutron has same kinetic energy, is passed through a sample of hydrogen like gas (but not hydrogen) in ground state. Due to collision of neutrons with the ions of the gas, ions are excited and then they emit photons. Six spectral lines are obtained in which one of the lines is of wavelength (6200/51) nm. What is the minimum possible value of kinetic energy of the neutrons for this to be possible. The mass of neutron and proton can be assumed to be nearly same. Find the answer in the form $25\alpha \times 10^{-2}\ eV$ and fill value of α.

PMP222

5. There are some atoms in ground energy level and rest of all in some upper energy level in a gas of identical hydrogen - like atoms. The atoms jump to higher energy state after absorbing photon of energy 1.89 eV and as a result emit radiations of 3 different photon energies. Find the principal quantum number of initially excited level :-

PMP129

6. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level $n = 4$ to $n = 2$ in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}\ eV$ where α is the numerical constant. Find the value of α.

[Take : $hc = 12420\ eV$-Å]

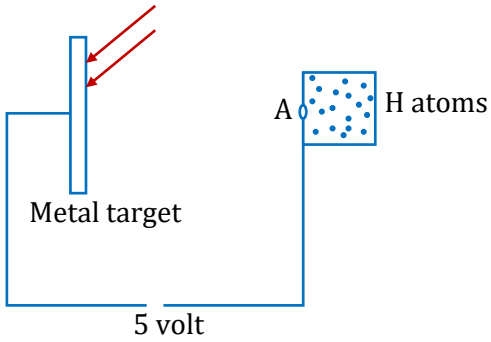
PMP221

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7. Monochromatic radiation emitted when electron on hydrogen atom jumps from first excited to the grounds state irradiates a photosensitive material. The stopping potential is measured to be 3.57 *V*. The threshold frequency of the material is $x \times 10^{14}$ *Hz*, then find value of *x*.

PMP139

8. Electromagnetic waves of wavelength 1242 Å are incident on a metal of work function 2 *eV*. The target metal is connected to a 5volt cell, as shown. The electrons pass through hole *A* into a gas of hydrogen atoms in their ground state. Find the number of spectral lines emitted when hydrogen atoms come back to their ground states after having been excited by the electrons. Assume all excitations in *H*-atoms from ground state only. (hc = 12420 *eV*Å)



PMP223

9. A He^+ ion in ground state is fired towards a Hydrogen atom in ground state and at rest. What should be the minimum kinetic energy (in *eV*) of He^+ ion so that both single electron species may get excited.

PMP224

10. A lithium atom has three electrons, Assume the following simple picture of the atom. Two electrons move close to the nucleus making up a spherical cloud around it and the third moves outside this cloud in a circular orbit. Bohr's model can be used for the motion of this third electron but n = 1 states are not available to it. Calculate the ionization energy of lithium in ground state using the above picture.

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PMP225

11. An imaginary particle has a charge equal to that of an electron and mass 100 times the mass of the electron. It moves in a circular orbit around a nucleus of charge + 4*e*. Take the mass of the nucleus to be infinite. Assuming that the Bohr's model is applicable to the system.

(a) Derive and expression for the radius of nth Bohr orbit.

(b) Find the wavelength of the radiation emitted when the particle jumps from fourth orbit to the second.

PMP227

12. An electron is orbiting in a circular orbit of radius *r* under the influence of a constant magnetic field of strength *B*. Assuming that Bohr's postulate regarding the quantisation of angular momentum holds good for this electron, find:

(a) the allowed values of the radius '*r*' of the orbit.

(b) the kinetic energy of the electron in orbit

(c) the potential energy of interaction between the magnetic moment of the orbital current due to the electron moving in its orbit and the magnetic field *B*.

(d) the total energy of the allowed energy levels.

PMP228

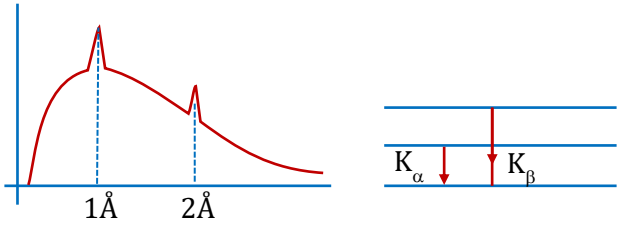
13. A neutron with an energy of 4.6 *MeV* collides with protons and is retarded. Assuming that upon each collision neutron is deflected by 45° find the number of collisions which will reduce its energy to 0.23 *eV*.

PMP229

For More Join: @IITJEE_Advanced

X-rays

14. Figure shows K_α & K_β *X*-rays along with continuous *X*-ray. Find the energy of L_α *X*-ray.
(Use $hc = 12420\ ev\text{Å}$).



PMP167

De Broglie

15. The potential energy of a particle varies as $U(x) = E_0$ for $0 < x < 1$ and $U(x) = 0$ for $x > 1$ for $0 \le x \le 1$. de–Broglie wavelengths is λ_1 and for $x > 1$ the de–Broglie wavelength is λ_2.

Total energy of the particle is $2E_0$. Find $\frac{\lambda_1}{\lambda_2}$.

PMP101

16. Two identical nonrelativistic particles move at right angles to each other, possessing De Broglie wavelengths, λ_1 and λ_2. Find the De Broglie wavelength of each particle in the frame of their centre of mass.

PMP102

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Nuclear Physics

17. In an α-decay the Kinetic energy of a particle is 48 *MeV* and *Q*-value of the reaction is 50 *MeV*. The mass number of the mother nucleus is: (Assume that daughter nucleus is in ground state)

PMP183

18. Consider the beta decay $^{198}Au \rightarrow {}^{198}Hg^* + \beta^- + \bar{\nu}$ where $^{198}Hg^*$ represents a mercury nucleus in an excited state at energy 1.088 *MeV* above the ground state. What can be the maximum kinetic energy of the electron emitted? The atomic mass ^{198}Au is 197.968233 *u* and that of ^{198}Hg is 197.966760 *u*.

PMP230

Radioactivity

19. U^{238} and U^{235} occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth.
(Half life of $u^{238} = 4.5 \times 10^9\ yrs$ & that of $U^{235} = 7.13 \times 10^8\ yrs$)

PMP209

20. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is $10^9 s$. The mass of an atom of this radioisotope is $10^{-25}\ kg$. The mass (in *mg*) of the radioactive sample is

PMP214

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EXERCISE - JEE (Main) PYQ

1. Radiation of wavelength λ, is incident on a photocell. The fastest emitted electron has speed v. If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be :-

[JEE (Main) 2016]

(1) $= v\left(\frac{3}{5}\right)^{1/2}$ (2) $> v\left(\frac{4}{3}\right)^{1/2}$ (3) $< v\left(\frac{4}{3}\right)^{1/2}$ (4) $= v\left(\frac{4}{3}\right)^{1/2}$

PMP036

2. A radioactive nucleus A with a half life T, decays into a nucleus B. At $t = 0$, there is no nucleus B. At sometime t, the ratio of the number of B to that of A is 0.3. Then, t is given by:

[JEE (Main) 2017]

(1) $t = T \log (1.3)$ (2) $t = \frac{T}{\log(1.3)}$ (3) $t = \frac{T}{2}\frac{\log 2}{\log 1.3}$ (4) $t = T\frac{\log 1.3}{\log 2}$

PMP039

3. An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the De-Broglie wavelength of the electron in the nth state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the nth state to the ground state. For large n, (A, B are constants)

[JEE (Main) 2018]

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(1) $\Lambda_n \approx A + B\lambda_n$ (2) $\Lambda_n^2 \approx A + B\lambda_n^2$ (3) $\Lambda_n^2 \approx \lambda$ (4) $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

PMP041

4. Half lives of two radioactive nuclei A and B are 10 minutes and 20 minutes, respectively. If initially a sample has equal number of nuclei, then after 60 minutes, the ratio of decayed numbers of nuclei A and B will be :

[JEE (Main) 2019]

(1) 9 : 8 (2) 1 : 8 (3) 8 : 1 (4) 3 : 8

PMP042

5. In a reactor, 2 kg of $^{235}_{92}U$ fuel is fully used up in 30 days. The energy released per fission is 200 *MeV*. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and 1 $eV = 1.6 \times 10^{-19}$ J. The power output of the reactor is close to :

[JEE (Main) 2020]

(1) 125 *MW* (2) 60 *MW* (3) 35 *MW* (4) 54 *MW*

PMP043

6. The De-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the De-Broglie wavelength reduces to 75% of the initial value ?

[JEE (Main) 2021]

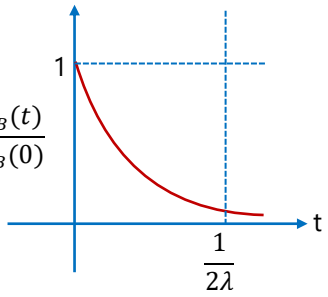
(1) $\frac{1}{9}E$ (2) $\frac{7}{9}E$ (3) E (4) $\frac{16}{9}E$

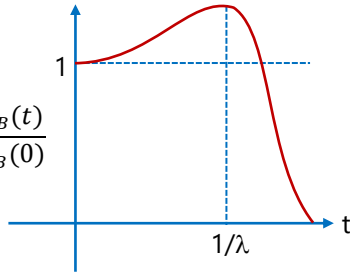
PMP045

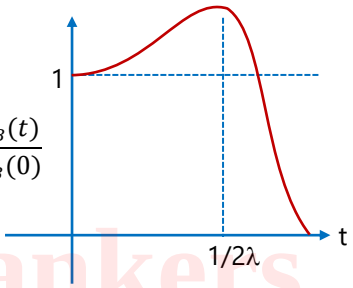
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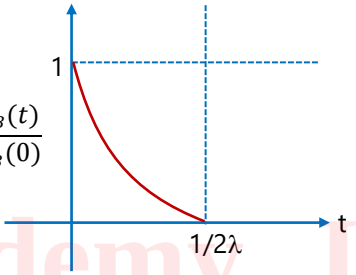
7. At time t = 0, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ. However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t)$ / $N_B(0)$ with respect to time t ? **[JEE (Main) 2021]**

$\begin{bmatrix} N_A(0) = \text{No. of } A \text{ atoms at } t = 0 \\ N_B(0) = \text{No. of } B \text{ atoms at } t = 0 \end{bmatrix}$

(1) 

(2) 

(3) 

(4) 

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PMP046

8. The light of two different frequencies whose photons have energies 3.8 *eV* and 1.4 *eV* respectively, illuminate a metallic surface whose work function is 0.6 *eV* successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be: **[JEE (Main) 2022]**

(1) 1 : 1 (2) 2 : 1 (3) 4 : 1 (4) 1 : 4

PMP047

9. In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level : **[JEE (Main) 2022]**

(1) All K, P and E increase
(2) K decreases, P and E increase
(3) P decreases, K and E increase
(4) K increases, P and E decrease

PMP048

10. A sample contains 10^{-2} kg each of two substances A and B with half lives 4s and 8s respectively. The ratio of then atomic weights is 1 : 2. The ratio of the amounts of A and B after 16s is $\frac{x}{100}$. The value of x is _____. **[JEE (Main) 2022]**

PMP049

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11. An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state, the energy released in this process, will be:
(Given $Rch = 13.6\ eV$)
Where R = Rydberg constant, c = Speed of light in vacuum, h = Planck's constant

[JEE (Main) 2023]

(1) 13.6 eV (2) 10.5 eV (3) 3.4 eV (4) 40.8 eV

PMP234

12. A radioactive nucleus decays by two different process. The half-life of the first process is 5 minutes and that of the second process is 30s. The effective half-life of the nucleus is calculated to be $\frac{\alpha}{11}$ s. The value of α is _____. **[JEE (Main) 2023]**

PMP235

13. An α-particle, a proton and an electron have the same kinetic energy. Which one of the following is correct in case of their De-Broglie wavelength? **[JEE (Main) 2023]**

(1) $\lambda_\alpha > \lambda_p > \lambda_e$ (2) $\lambda_\alpha < \lambda_p < \lambda_e$ (3) $\lambda_\alpha = \lambda_p = \lambda_e$ (4) $\lambda_\alpha > \lambda_p < \lambda_e$

PMP236

14. From the photoelectric effect experiment, following observations are made. Identify which of these are correct

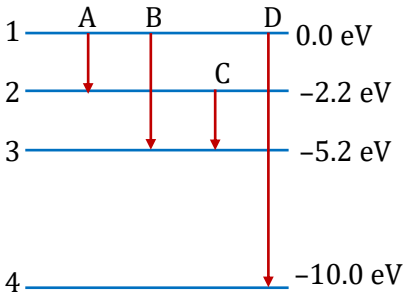
A. The stopping potential depends only on the work function of the metal.
B. The saturation current increases as the intensity of incident light increases.
C. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.
D. Photoelectric effect can be explained using wave theory of light.

Choose the correct answer from the options given below: **[JEE (Main) 2023]**

(1) B, C only (2) A, C, D only (3) B only (4) A, B, D only

PMP237

15. The energy levels of an atom is shown is figure. Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm ? Given ($h = 6.62 \times 10^{-34}\ Js$)



[JEE (Main) 2023]

(1) B (2) A (3) C (4) D

PMP238

16. The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

[JEE (Main) 2024]

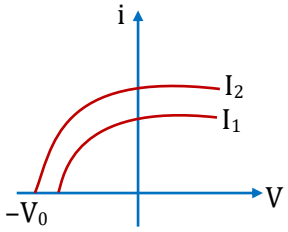
(1) $\frac{1}{1}$ (2) $\frac{1}{8}$ (3) $\frac{8}{1}$ (4) $\frac{1}{4}$

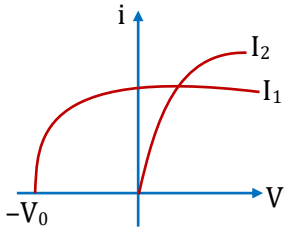
PMP256

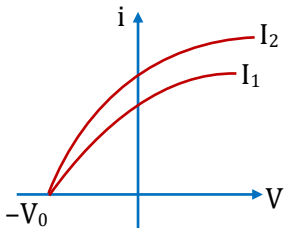
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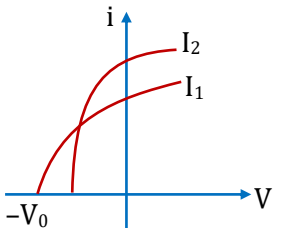
17. Which figure shows the correct variation of applied potential difference (V) with photoelectric current (I) at two different intensities of light ($I_1 < I_2$) of same wavelengths :

[JEE (Main) 2024]

(1) 

(2) 

(3) 

(4) 

PMP257

18. Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Number of photons increases with increase in frequency of light.

Reason R : Maximum kinetic energy of emitted electrons increases with the frequency of incident radiation.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

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[JEE (Main) 2024]

(1) Both **A** and **R** are correct and **R** is **NOT** the correct explanation of **A**.

(2) **A** is correct but **R** is not correct.

(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(4) **A** is not correct but **R** is correct.

PMP258

19. An electron of mass 'm' with an initial velocity $\vec{v} = v_0\hat{i}(v_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{k}$. If the initial de Broglie wavelength is λ_0, the value after time t would be :- **[JEE (Main) 2025]**

(1) $\dfrac{\lambda_0}{\sqrt{1+\dfrac{e^2E_0^2t^2}{m^2v_0^2}}}$ (2) $\dfrac{\lambda_0}{\sqrt{1-\dfrac{e^2E_0^2t^2}{m^2v_0^2}}}$ (3) λ_0 (4) $\lambda_0\sqrt{1+\dfrac{e^2E_0^2t^2}{m^2v_0^2}}$

PMP302

20. During the transition of electron from state A to state C of a Bohr atom, the wavelength of emitted radiation is 2000 Å and it becomes 6000 Å when the electron jumps from state B to state C. Then the wavelength of the radiation emitted during the transition of electrons from state A to state B is:

[JEE (Main) 2025]

(1) 3000 Å (2) 6000 Å (3) 4000 Å (4) 2000 Å

PMP303

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EXERCISE - JEE (Advanced) PYQ

1. Match the nuclear processes given in column-I with the appropriate option(s) in column-II.

[JEE (Advanced) 2015]

	Column-I		**Column-II**
(A)	Nuclear fusion	(P)	Absorption of thermal neutrons by $^{235}_{92}U$
(B)	Fission in a nuclear reactor	(Q)	$^{60}_{27}Co$ nucleus
(C)	β-decay	(R)	Energy production in stars via hydrogen conversion to helium
(D)	γ-ray emission	(S)	Heavy water
		(T)	Neutrino emission

PMP076

2. The isotope $^{12}_{5}B$ having a mass 12.014 u undergoes β-decay to $^{12}_{6}C \cdot {}^{12}_{6}C$ has an excited state of the nucleus ($^{12}_{6}C^*$) at 4.041 $M\,eV$ above its ground state. If $^{12}_{5}B$ decays to $^{12}_{6}C^*$, the maximum kinetic energy of the β-particle in units of $M\,eV$ is (1u = 931.5 $M\,eV/c^2$, where c is the speed of light in vacuum). **[JEE (Advanced) 2016]**

PMP077

3. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of radius R is given by $E = \frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$. The measured masses of the neutron, $^{1}_{1}H$, $^{15}_{7}N$ and $^{15}_{8}O$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the $^{15}_{7}N$ and $^{15}_{8}O$ nuclei are same, $1u = 931.5\ M\,eV/c^2$ (c is the speed of light) and $\frac{e^2}{(4\pi\varepsilon_0)} = 1.44\ M\,eV\,fm$. Assuming that the difference between the binding energies of $^{15}_{7}N$ and $^{15}_{8}O$ is purely due to the electrostatic energy, the radius of either of the nuclei is ($1fm = 10^{-15}m$)

[JEE (Advanced) 2016]

(A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

PMP078

4. An electron in a hydrogen atom undergoes a transition from an orbit with quantum number n_i to another with quantum number n_f. V_i and V_f are respectively the initial and final potential energies of the electron. If $\frac{V_i}{V_f} = 6.25$, then the smallest possible n_f is. **[JEE (Advanced) 2017]**

PMP079

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5. In a radioactive decay chain, $^{232}_{90}Th$ nucleus decays to $^{212}_{82}Pb$ nucleus. Let N_α and N_β be the number of α and β^- particles, respectively, emitted in this decay process. Which of the following statements is (are) true ? **[JEE (Advanced) 2018]**

(A) $N_\alpha = 5$ (B) $N_\alpha = 6$ (C) $N_\beta = 2$ (D) $N_\beta = 4$

PMP080

6. Suppose a $^{226}_{88}Ra$ nucleus at rest and in ground state undergoes α-decay to a nucleus in its excited state. The kinetic energy of the emitted α particle is found to be 4.44 MeV. $^{222}_{86}Rn$ nucleus then goes to its ground state by γ-decay. The energy of the emitted γ-photon is _____ keV.

[Given: atomic mass of $^{226}_{88}Ra = 226.005u$, atomic mass of $^{222}_{86}Rn = 222.000u$, atomic mass of α particle = $4.000u$, $1u$ = 931 MeV/c^2, c is speed of the light]

[JEE (Advanced) 2019]

PMP081

7. In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{I}{2}$, the potential difference V is increased to $2V$ and the separation distance d is reduced to $\frac{d}{2}$, then

[JEE (Advanced) 2020]

(A) the cut-off wavelength will reduce to half and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

PMP082

8. A heavy nucleus Q of half-life 20 minutes undergoes alpha-decay with probability of 60% and beta-decay with probability of 40%. Initially, the number of Q nuclei is 1000. The number of alpha-decays of Q in the first one hour is **[JEE (Advanced) 2021]**

(A) 50 (B) 75 (C) 350 (D) 525

PMP083

9. In a photoemission experiment, the maximum kinetic energies of photoelectrons from metals P, Q and R are E_P, E_Q and E_R respectively and they are related by $E_P = 2E_Q = 2E_R$. In this experiment, the same source of monochromatic light is used for metals P and Q while a different source of monochromatic light is used for the metal R. The work functions for metals P, Q and R are 4.0 eV, 4.5 eV and 5.5 eV respectively. The energy of the incident photon used for metal R, in eV, is _____.

[JEE (Advanced) 2021]

PMP084

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10. The minimum kinetic energy needed by an alpha particle to cause the nuclear reaction ${}^{16}_{7}N + {}^{4}_{2}He \rightarrow {}^{1}_{1}H + {}^{19}_{8}O$ in a laboratory frame is n (in MeV). Assume that ${}^{16}_{7}N$ is at rest in the laboratory frame. The masses of ${}^{16}_{7}N$, ${}^{4}_{2}He$, ${}^{1}_{1}H$ and ${}^{19}_{8}O$ can be taken to be 16.006 u, 4.003 u, 1.008 u and 19.003 u, respectively, where 1 u = 930 $MeVc^{-2}$. The value of n is_____.

[JEE (Advanced) 2022]

PMP085

11. The binding energy of nucleons in a nucleus can be affected by the pairwise Coulomb repulsion. Assume that all nucleons are uniformly distributed inside the nucleus. Let the binding energy of a proton be E_b^p and the binding energy of a neutron be E_b^n in the nucleus. Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2022]**

(A) $E_b^p - E_b^n$ is proportional to $Z(Z-1)$ where Z is the atomic number of the nucleus.

(B) $E_b^p - E_b^n$ is proportional to $A^{\frac{-1}{3}}$ where A is the mass number of the nucleus.

(C) $E_b^p - E_b^n$ is positive.

(D) E_b^p increases if the nucleus undergoes a beta decay emitting a positron.

PMP086

12. In a radioactive decay chain reaction, ${}^{230}_{90}Th$ nucleus decays into ${}^{214}_{84}Po$ nucleus. The ratio of the number of α to number of β^- particles emitted in this process is _____.

[JEE (Advanced) 2022]

PMP087

13. A Hydrogen-like atom has atomic number Z. Photons emitted in the electronic transitions from level n = 4 to level n = 3 in these atoms are used to perform photoelectric effect experiment on a target metal. The maximum kinetic energy of the photoelectrons generated is 1.95 eV. If the photoelectric threshold wavelength for the target metal is 310 nm, the value of Z is _____.

[Given: hc = 1240 eV-nm and Rhc = 13.6 eV, where R is the Rydberg constant, h is the Planck's constant and c is the speed of light in vacuum] **[JEE (Advanced) 2023]**

PMP239

14. In a radioactive decay process, the activity is defined as $A = -\frac{dN}{dt}$, where $N(t)$ is the number of radioactive nuclei at time t. Two radioactive sources, S_1 and S_2 have same activity at time t = 0. At a later time, the activities of S_1 and S_2 are A_1 and A_2, respectively. When S_1 and S_2 have just completed their 3rd and 7th half-lives, respectively, the ratio A_1/A_2 is _____. **[JEE (Advanced) 2023]**

PMP240

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15. A particle of mass m is moving in a circular orbit under the influence of the central force $F(r) = -kr$, corresponding to the potential energy $V(r) = kr^2 / 2$, where k is a positive force constant and r is the radial distance from the origin. According to the Bohr's quantization rule, the angular momentum of the particle is given by $L = n\hbar$, where $\hbar = h/(2\pi)$, h is the Planck's constant, and n a positive integer. If v and E are the speed and total energy of the particle, respectively, then which of the following expression(s) is(are) correct?

[JEE (Advanced) 2024]

(A) $r^2 = n\hbar\sqrt{\frac{1}{mk}}$ (B) $v^2 = n\hbar\sqrt{\frac{k}{m^3}}$ (C) $\frac{L}{mr^2} = \sqrt{\frac{k}{m}}$ (D) $E = \frac{n\hbar}{2}\sqrt{\frac{k}{m}}$

PMP259

16. A metal target with atomic number $Z = 46$ is bombarded with a high energy electron beam. The emission of X-rays from the target is analyzed. The ratio r of the wavelengths of the K_α-line and the cut-off is found to be $r = 2$. If the same electron beam bombards another metal target with $Z = 41$, the value of r will be

[JEE (Advanced) 2024]

(A) 2.53 (B) 1.27 (C) 2.24 (D) 1.58

PMP260

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	B	B	A	B	D	C	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	A	A	C	A	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	D	B	A	C	D	C	B	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	A	A	D	B	A	A	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	B	C	A	B	A	C	C	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	C	A	B	B	D	C	A	C	C
Que.	61	62								
Ans.	B	C								

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EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABCD	BCD	ABC	ABD	BCD	AC	AB	ABD	BC	ABCD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	BD	ACD	ACD	BC	ACD	AC	ACD	CD	BD	ABC
Que.	21									
Ans.	A									

EXERCISE (S)

1. 4 **2.** 5 **3.** 1 **4.** 255

5. 2 **6.** 159 **7.** 16 **8.** 6

9. 255 **10.** 3.4 eV **11.** (a) $\frac{n^2h^2\varepsilon_0}{400\ \pi me^2}$ (b) 3.0 $\AA$

12. (a) $\sqrt{\frac{nh}{2\pi Be}}$ (b) $\frac{nhBe}{4\pi m}$ (c) $\frac{nheB}{4\pi m}$ (d) $\frac{nheB}{2\pi m}$ **13.** 24 **14.** 6210 eV

15. $\sqrt{2}$ **16.** $\lambda = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2+\lambda_2^2}}$ **17.** 100 **18.** 0.2806 MeV

19. $6.04 \times 10^9\ yrs$ **20.** 1

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	4	1	2	2	3	2	2	50
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	300	2	3	4	2	3	4	1	1

EXERCISE - JEE (Advanced) PYQ

Que.	1			2	3	4	5	6	7
Ans.	(A-R or R,T), (B-P,S), (C-Q,T), (D-R)			9	C	5	AC	135	AC
Que.	8	9	10	11	12	13	14	15	16
Ans.	D	6	2.32 to 2.33	ABD	2	3	16	ABC	A

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